Front End Engineering

Front End Engineering II evaluation- 2

(Batch-2025)

PheW TuBe (Video Streaming Platform)



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Evaluation-II Vision & Milestone Objectives

Phase 1 Foundation vs Phase 2 Focus

- **Phase 1 Achievement:** Established static catalog, card layouts, sidebar navigation, and external routing.
- **Phase 2 Transition:** Elevating PheW TuBe into an authentic, state-driven Single Page Application (SPA).
- **Zero Page Reloads:** Full SPA transition utilizing client-side routing, virtual DOM reconciliation, and instant state reactivity.

Core Phase 2 Technical Objectives

- **Dynamic Watch Routing:** Deep link `/watch/:id` powered by `useParams()` with embedded 16:9 streaming.
- **DOM Control with useRef:** Direct programmatic focus on comment input and auto-scroll viewport control without synthetic re-renders.
- **Global Context API:** Centralized multi-state bus eliminating prop drilling across nested component hierarchies.

User Experience & Realism Goals

- **Up-Next Recommendations:** Dynamic video swapping via `useNavigate()` with synchronized metadata.
- **Interactive Notification Bus:** Integrated `react-toastify` for instant feedback on user actions.
- **Responsive Parity:** Seamless layout scaling across 375px mobile, 768px tablet, and 1440px desktop screens.

Syllabus-Aligned Technology Stack & Architecture

React 19 & Component Architecture

- **Functional Components:** 100% modular, pure functional architecture with explicit hook lifecycles.
- **React Router DOM v6:** Declarative routing engine featuring parameterized dynamic routes (`:id`).
- **Vite 8 Build Tooling:** Lightning fast ESM bundling delivering sub-350ms hot module replacement (HMR).

Hooks & Memory Lifecycle Layer

- **useState & useEffect:** Controlled form inputs, mount lifecycle triggers, and document title synchronization.
- **useRef Hook:** Mutable non-rendering container for DOM focus control and debounced keyboard listeners.
- **useParams & useNavigate:** URL segment parsing and programmatic navigation without full page reloads.

State Persistence & UX Libraries

- **Context API (createContext):** Shared application state for library, history, subscriptions, and auth.
- **Browser LocalStorage:** Client-side JSON persistence ensuring data survives page refreshes and browser restarts.
- **react-toastify:** Professional non-blocking alert toasts recommended by course supervisor Mr. Shubham Kumar.

Core Component Architecture & Modularity

Component Hierarchy & Single Responsibility Principle

- **<AppProvider> (Root Context):** Houses global state slices, reducer actions, and localStorage side effects.
- **<Navbar>:** Independent search bar with live auto-complete dropdown, microphone voice trigger, and auth profile modal.
- **<Sidebar> & <CategoryChips>:** Two-way bound navigation and category filtering driving video feeds.
- **<VideoCard> & <ShortsCard>:** Pure presentation components passing IDs to dynamic watch routes via navigate hooks.
- **<WatchPage>:** Master composite page integrating video player, channel actions, comments feed, and recommended sidebar.

Data Flow & Isolation Architecture

- **Unidirectional Data Flow:** Context state descends via hooks; child actions bubble up via context methods (`toggleLike` , `addComment`).
- **Prop-Drilling Elimination:** WatchPage, Navbar, and Sidebar independently consume `useApp()` without intermediary prop relaying.
- **Modular Route Segregation:** Distinct routes for `/` , `/watch/:id` , `/shorts` , `/history` , `/liked` , `/watch-later` , and `/subscriptions` .
- **Authentication Modal Isolation:** Client-side auth popup controlled by Context state, avoiding unneeded DOM re-renders in parent pages.

Dynamic Watch Route & Embedded Video Engine (/watch/:id)

URL Parameter Extraction via useParams()

- **Dynamic Route Definition:** Configured in App.jsx as `<Route path="/watch/:id" />`.
- **Runtime Extraction:** `const { id } = useParams()` instantly queries dataset without page reloads.
- **Graceful Fallback:** Fallback screen renders with redirect button if invalid or non-existent video ID is supplied.

Embedded 16:9 Responsive Streaming Engine

- **Native YouTube Embed:** Embeds legitimate YouTube stream IDs with autoplay, picture-in-picture, and fullscreen flags.
- **Auto-Scroll on Route Switch:** `window.scrollTo(0, 0)` fires in `useEffect` on `[id]` change for smooth navigation.
- **History Auto-Recording:** Automatically pushes viewed video into persistent Watch History via Context API.

Up-Next Recommendation Engine

- **Category-Aware Sidebar:** Automatically filters catalog to present related videos in right-hand column.
- **Seamless Swap:** Clicking recommendation triggers `useNavigate('/watch/' + nextId)` instantly swapping player state.
- **Interactive Action Bar:** Interactive Like, Watch Later, and Clipboard Share with instant toast feedback.

Interactive Comments System with useRef & Dynamic Sorting

Targeted DOM Access via useRef Hook

- **Direct Input Referencing:** Initialized via `const commentInputRef = useRef(null)` and bound to input element.
- **Programmatic Focus:** Calling `commentInputRef.current.focus()` focuses comment box without triggering component re-rendering.
- **Controlled State Binding:** Controlled input state (`useState`) manages character text, clearing cleanly upon submission.
- **Defensible Viva Design:** Strategic application of useRef demonstrates genuine understanding of DOM interaction vs state mutations.

Multi-Video Comment Isolation & Sorting

- **Isolated Video Keys:** Comments mapped in state dictionary by video ID: `comments[videoId] = [...]`.
- **Dynamic Sorting Engine:** Users toggle between **Top Comments** (sorted by upvote count) and **Newest First** (chronological order).
- **Optimistic Updates:** Submitted comments appear instantly at the head of the feed with "Just now" timestamp.
- **Persistent Storage:** New comments persist to browser `localStorage`, retaining discussion threads across visits.

Global State Architecture (Context API) & LocalStorage Sync

Context API Design (AppContext.jsx)

- **Single Source of Truth:** Centralizes state for videos, shorts, likes, watch later, history, subscriptions, and comments.
- **Custom Hook Convenience:** Encapsulates consumer logic into `useApp()`, throwing errors if accessed outside Provider.
- **High Modularity:** Decouples UI rendering from business logic and data mutations.

Bidirectional LocalStorage Synchronization

- **Lazy State Initializers:** State initializes directly from `localStorage.getItem()` with safe fallback defaults.
- **Reactive Serialization:** Dedicated `useEffect` listeners automatically serialize state changes to browser storage.
- **Offline Data Durability:** Liked videos, watch history, and user settings persist across page reloads and browser sessions.

Client-Side Authentication Pattern

- **Curriculum-Aligned Auth:** Implements teacher-recommended modal sign-in/sign-up pattern (ecom class architecture).
- **Session State:** Tracks user name, university email, and dynamic avatar status in persistent state.
- **Access Validation:** Displays toast alerts prompting sign-in when attempting protected actions like upload or comment.

Creator Ecosystem & Exact 2026 YouTube Metrics

Global Megastars (Entertainment & Gaming)

- **MrBeast: 522M subscribers** — YouTube's #1 creator; features 7-Days Stranded and Private Island challenges.
- **T-Series: 316M subscribers** — World's leading Indian music label; features viral soundtrack releases.
- **Techno Gamerz (Ujjwal): 53.4M subscribers** — India's gaming icon; features GTA 5 Episode 147 and Minecraft survival.

Comedy & Specialized Gaming Creators

- **Samay Raina: 7.65M subscribers** — Standup comedy & India's Got Latent episodes featuring top guest panels.
- **Wemmbu: 3.45M subscribers** — Hardcore Minecraft survival, traps, and secret base architectural walkthroughs.
- **Beyond Coding: 415K subscribers** — Engineering leadership, system design, and software architecture interviews.

Syllabus & Developer Education Channels

- **freeCodeCamp.org: 12.0M subscribers** — Industry standard open curriculum; features comprehensive 10-Hour DSA course.
- **Chai aur Code (Hitesh): 645K subscribers** — Hindi developer education; features foundational React 19 series.
- **NeetCode / Striver: 980K subscribers** — Data structures, competitive algorithms, and tech interview masterclasses.

Multi-Device Responsive Design & Modern UI/UX Layer

Responsive Breakpoints Architecture

- **Desktop (>1150px):** 4-column video feed, collapsible full sidebar, dual-column Watch page with sidebar queue.
- **Tablet (768px - 1150px):** 2-3 column fluid grid, compact icon sidebar, stacked Watch page layout.
- **Mobile (<480px):** Single column full-bleed cards, bottom navigation parity, full-width responsive video player.

Live Search Autocomplete Engine

- **Instant Substring Matching:** Filters video titles, creator names, and tags in real-time as the user types.
- **Floating Dropdown Menu:** Rendered with subtle shadow elevation, keyboard accessibility, and outside-click dismiss.
- **Direct Video Launch:** Selecting any suggestion navigates straight to that video's dynamic player route.

Micro-Interactions & Toast Bus

- **react-toastify Alerts:** Dark-themed feedback badges on likes, playlist additions, and clipboard copy.
- **Shorts Feed Scroll-Snap:** CSS `scroll-snap-type: y mandatory` for smooth vertical reel browsing.
- **Active State Feedback:** Red heart fills, pill background changes, and animated voice microphone pulse rings.

Curriculum Alignment & Faculty Reference Verification

Class Reference Alignment (Mr. Shubham Kumar)

- **Reference Codebase Inspection:** Validated against faculty GitHub repository (`shubhamkumar123456/G15-updated`).
- **Course Hook Implementations:** Fully implements `useParams` , `useRef` , `useEffect` , and `useState` in accordance with semester syllabus.
- **State Architecture:** Context API and persistent LocalStorage follow faculty architectural patterns demonstrated in class.
- **Faculty Recommendations:** Incorporated `react-toastify` as explicitly recommended by faculty during lab lectures.

Evaluation-II Viva Readiness & Proofs

- **Strict Production Separation:** Private viva preparation guides and scratch notes kept strictly local; only production code pushed to GitHub.
- **Measurable Metrics:** Zero external redirect bounces; 100% in-app streaming; 16 videos, 8 shorts across 10 categories.
- **Component Explainability:** Code written with clean modularity, standard variable naming, and clear functional boundaries for confident viva defense.
- **Standard File Naming:** Output deck formatted strictly as `G15-PID-17.pdf` for formal portal submission.

Technical Roadmap & Scope for Phase 3 (MERN Stack)

Phase 1: Foundation (Completed)

- Complete UI Layout & Dark Theme
- Static Video & Shorts Grid
- Multi-Category Filtering Pills
- Static Sidebar Navigation
- Basic Search Query Input

Phase 2: SPA Architecture (Current)

- Dynamic Route: `/watch/:id` via useParams
- Embedded 16:9 YouTube Streaming Player
- Interactive Comments with useRef
- Centralized Context API & LocalStorage
- Live Autocomplete & react-toastify Alerts
- Full Laptop, Tablet, & Mobile Responsiveness

Phase 3: Full Stack Scaling (Roadmap)

- **Node.js & Express API:** RESTful backend endpoints replacing mock JSON dataset.
- **MongoDB & Mongoose:** Cloud database storing user profiles, videos, and comments.
- **JWT & Bcrypt Security:** Server-verified authentication with encrypted user tokens.
- **Redux Toolkit & RTK Query:** Advanced cached state management and optimistic mutations.